

CSC 309

Data Structure and Algorithm

Examination: Semester Examination	Semester: Second Semester
Year: 2023	Duration: 2 hours
Total Questions: 6	

Instructions: Answer ALL questions. All questions carry equal marks unless otherwise stated.

Question 1

- (a) What is a queue? Explain the enqueue and dequeue operations. (6 marks)
- (b) Differentiate between linear queue and circular queue. (5 marks)
- (c) Write an algorithm to implement a queue using two stacks. (4 marks)

Question 2

- (a) Explain Dijkstra's shortest path algorithm with a suitable example. (10 marks)
- (b) What is the time complexity of Dijkstra's algorithm? When does it fail? (5 marks)

— End of Examination —